

3.1 Equipment

Item	Requirement	Why
VSX3000 sensor on a rigid stand	Stand mechanically separate from the robot base, on an isolated floor or table	Prevents robot motion from moving the sensor
6-axis robot	Repeatability ≤ 0.05 mm (ISO 9283); payload above the heaviest target plus adapter	Commanded, repeatable target poses
Quick-change target adapter	Dowel-pinned, re-mount repeatability ≤ 0.02 mm	Targets can be swapped without re-registering
Enclosure or blackout curtains, IR light meter	Ambient IR held constant and logged	Ambient IR changes the noise level
Temperature loggers (sensor housing and air)	1 sample/min	For drift attribution
Capture PC	VSX3000 SDK, LRVisionLibs MatCloud reader, robot interface that logs the actual pose read back from the robot	Synchronized capture and logging

3.2 Targets

ID	Target	Construction	Used in
T1	Registration plate	Flat, about 400 × 300 mm, with a ChArUco or circle-grid pattern that is visible in the left IR image. Pattern geometry is known to ≤ 0.02 mm.	4
T2	Noise plate	Uniform matte, NOISE_PLATE_SIZE_MM, flatness PLATE_FLATNESS_MM	A, B-Z, sentinels
T3a	Raised square	Square of side EDGE_SQUARE_SIZE_MM with knife edges, on hidden posts at G above a back plate. Spacers set G to GAP_SMALL_MM or GAP_LARGE_MM.	B-HV, E
T3b	Square window	Front plate with a square window of the same size and knife edges; back plate at G behind it	B-HV, E
T4-S / T4-L	Disk arrays, small and large	Back-beveled disks on thin posts at G above a back plate. The diameter ladder is split across two plates by size. Includes blank sites and post-only control sites.	C, D, E
T5-S / T5-L	Cutout arrays, small and large	Back-beveled holes in a front plate, with the back plate at G. Includes blank sites.	C, D, E

Diameter ladder. Diameters run from $\text{DIAMETER_MIN_FOOTPRINT_FRACTION} \times p(Z_MIN)$ up to $\text{DIAMETER_MAX_FOOTPRINT_MULTIPLE} \times p(Z_MAX)$, in steps of $\text{DIAMETER_LADDER_RATIO}$. For example, at $f_x \approx 500$ px (unconfirmed) this gives about 0.3 mm to 60 mm in about 16 steps. Features on a plate are spaced at least $\text{FEATURE_ISOLATION_PX}$ apart at Z_MIN , so the sensor's spatial interpolation cannot couple neighboring features.

Blank and control sites. Each array has blank sites matching each search-window size. These give the false-alarm rate in D. Disk arrays also have post-only sites (a post with no disk). These show whether the support post itself is detected.

Surface finish. All front and back surfaces use one finish, for example bead-blasted aluminum or a matte coating, with mid-range IR reflectance. Record the finish and, if possible, its reflectance at the projector wavelength. A reflectance difference between plates would bias the edge and area results.